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1. Executive Summary

This is a non-technical introduction to the quantum threat, NIST's post-quantum cryptography standards, and the practical steps organisations should take today to prepare for the transition.

Quantum computing has the potential to break many of the cryptographic systems currently used to secure sensitive data, communications, and digital identities.

While practical quantum attacks are not yet mainstream, organisations must begin preparing now because cryptographic migration programmes can take years to complete.

This guide explains the quantum threat in business terms, introduces the emerging post-quantum cryptography (PQC) standards published by the U.S. National Institute of Standards and Technology (NIST), and outlines a practical readiness strategy for decision-makers.

2. What Is the Quantum Threat?

Modern encryption systems rely on mathematical problems that are extremely difficult for classical computers to solve.

Quantum computers use fundamentally different processing methods that may allow them to solve some of these problems significantly faster.

If large-scale fault-tolerant quantum computers become viable, widely used encryption systems such as RSA and elliptic curve cryptography (ECC) could become vulnerable.

3. Why This Matters to Organisations

The quantum threat is not only a future problem. Threat actors may already be collecting encrypted data today with the intention of decrypting it later once quantum capabilities mature. This is often referred to as 'harvest now, decrypt later'.

Organisations with long-lived sensitive information are particularly exposed, including:

- Financial institutions
- Healthcare providers
- Government agencies
- Critical infrastructure operators
- Legal and professional services firms
- Organisations storing intellectual property or trade secrets

4. Which Technologies Are at Risk?

Quantum attacks primarily threaten public-key cryptography systems used for:

- Secure websites (TLS/SSL certificates)
- VPNs and remote access systems
- Digital signatures
- Identity and access management systems
- Email encryption
- Software signing and firmware validation
- Blockchain and cryptocurrency technologies

Symmetric encryption systems such as AES are less severely affected but may still require stronger key sizes.

5. What Is Post-Quantum Cryptography (PQC)?

Post-quantum cryptography refers to cryptographic algorithms designed to resist attacks from both classical and quantum computers.

Unlike quantum cryptography, PQC does not require specialised quantum hardware.

These algorithms can run on conventional systems and are intended to replace vulnerable public-key cryptographic methods.

6. NIST and the New PQC Standards

NIST has led a global multi-year effort to evaluate and standardise post-quantum cryptographic algorithms.

Key approved or selected algorithms include:

- CRYSTALS-Kyber (key encapsulation / encryption)
- CRYSTALS-Dilithium (digital signatures)
- SPHINCS+ (stateless hash-based signatures)
- FALCON (compact digital signatures)

These standards are expected to form the foundation of future cryptographic implementations across industries.

7. Why Organisations Must Start Preparing Now

Cryptographic transformation is complex and often deeply embedded across infrastructure, applications, devices, third-party integrations, and operational processes.

Large organisations may require several years to:

- Identify vulnerable cryptographic assets
- Upgrade legacy systems
- Replace unsupported applications
- Validate interoperability
- Update governance and policies
- Train technical teams

8. Common Organisational Challenges

Many organisations do not have visibility into where cryptography is used across their environment.

Common barriers include:

- Legacy applications with hardcoded encryption methods
- Unsupported or end-of-life systems
- Third-party vendor dependencies
- Lack of cryptographic inventory management
- Operational risk associated with replacing security controls

9. Recommended Actions for Decision-Makers

Organisations should begin structured preparation immediately.

Recommended actions include:

1. Establish executive awareness and governance ownership.
2. Create an inventory of cryptographic systems and dependencies.
3. Identify systems using vulnerable public-key cryptography.
4. Assess third-party supplier and vendor readiness.
5. Prioritise systems containing long-life sensitive data.
6. Develop a phased cryptographic migration strategy.
7. Monitor regulatory and standards developments.
8. Ensure future procurement requirements include PQC readiness.

10. Questions Boards and Executives Should Ask

Leadership teams should ask:

- Do we know where cryptography is used across the organisation?
- Which systems would be hardest to replace or upgrade?
- What data would still be sensitive in 10 years?
- Are our vendors preparing for post-quantum migration?
- Who owns quantum-readiness planning internally?
- Have we included PQC in technology roadmaps and procurement standards?

11. Regulatory and Industry Developments

Governments and standards bodies globally are beginning to issue quantum-readiness guidance.

Organisations operating in regulated industries should expect future compliance requirements related to cryptographic agility and quantum-safe migration planning.

Early preparation reduces future operational disruption and compliance risk.

12. Building Cryptographic Agility

Cryptographic agility refers to the ability to rapidly replace or update cryptographic algorithms without major operational disruption.

Organisations should prioritise architectures that allow cryptographic components to be upgraded without requiring complete system replacement.

13. Immediate Priorities for the Next 12 Months

Suggested near-term roadmap:

- Conduct a cryptographic discovery assessment
- Identify high-risk systems and sensitive data stores
- Engage strategic vendors regarding PQC readiness
- Update security architecture standards
- Include PQC readiness in procurement processes
- Develop executive reporting on quantum readiness risk

14. Conclusion

The transition to post-quantum cryptography is likely to become one of the largest security modernisation initiatives organisations will face over the next decade.

While the exact timeline for practical quantum attacks remains uncertain, the operational lead time required for migration means organisations should begin preparing now.

The most resilient organisations will treat post-quantum readiness as a strategic risk management and technology governance initiative rather than purely a technical exercise.

Appendix A – PQC Readiness Checklist

Use the following checklist as a high-level readiness assessment:

- ☐ Executive awareness of quantum-related risk established
- ☐ Ownership assigned for PQC readiness programme
- ☐ Cryptographic inventory initiated
- ☐ High-risk systems identified
- ☐ Vendor readiness assessments underway
- ☐ Procurement standards updated
- ☐ Long-life sensitive data classified
- ☐ Migration roadmap drafted
- ☐ Cryptographic agility requirements defined
- ☐ Ongoing monitoring of NIST and industry developments established

Contact Us

If you would like to discuss your organisation's specific requirements, risks, compliance obligations, cybersecurity posture, business continuity strategy, or operational resilience objectives in greater detail, we invite you to contact Montana Data Company.

Our team can assist with tailored assessments, strategic guidance, governance support, and practical implementation aligned to your organisation's unique operational and regulatory environment.

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